

Newton-Cotes Formulas of Closed Type

Example 06-01: Estimate the definite integral $\int_0^{1.8} \sqrt{1+x^2} \cdot dx$ using Newton-Cotes formulas of closed type of order 1,2,3 with 12 uniform subintervals.

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i	x_i	$f(x_i)$
0	0.00	1.000000
1	0.15	1.011187
2	0.30	1.044031
3	0.45	1.096586
4	0.60	1.166190
5	0.75	1.250000
6	0.90	1.345362
7	1.05	1.450000
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Order 1 $\rightarrow n = 1$

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Order 1 $\rightarrow n = 1 \rightarrow$ The Trapezoidal Rule

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Order 1 $\rightarrow n = 1 \rightarrow$ The Trapezoidal Rule 12 times

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Order 1 $\rightarrow n = 1 \rightarrow$ The Trapezoidal Rule **12 times**

$$\int_0^{1.8} \sqrt{1+x^2} \cdot dx = \int_0^{0.15} \sqrt{1+x^2} \cdot dx + \int_{0.15}^{0.30} \sqrt{1+x^2} \cdot dx + \dots + \int_{1.65}^{1.8} \sqrt{1+x^2} \cdot dx$$

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①

Order 1 $\rightarrow n = 1 \rightarrow$ The Trapezoidal Rule **12 times**

$$\int_0^{1.8} \sqrt{1+x^2} \cdot dx = \int_0^{0.15} \sqrt{1+x^2} \cdot dx + \int_{0.15}^{0.30} \sqrt{1+x^2} \cdot dx + \dots + \int_{1.65}^{1.8} \sqrt{1+x^2} \cdot dx$$

$$\begin{aligned} \int_{0.00}^{0.15} \sqrt{1+x^2} \cdot dx &\approx \frac{h}{2} \cdot [f(0.00) + f(0.15)] \\ &= \frac{0.15}{2} \cdot [1.000000 + 1.011187] \\ &= 0.150839 \end{aligned}$$

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②

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$$\int_0^{1.8} \sqrt{1+x^2} \cdot dx = \int_0^{0.15} \sqrt{1+x^2} \cdot dx + \int_{0.15}^{0.30} \sqrt{1+x^2} \cdot dx + \dots + \int_{1.65}^{1.8} \sqrt{1+x^2} \cdot dx$$

$$\begin{aligned} \int_{0.15}^{0.30} \sqrt{1+x^2} \cdot dx &\approx \frac{h}{2} \cdot [f(0.15) + f(0.30)] \\ &= \frac{0.15}{2} \cdot [1.011187 + 1.044031] \end{aligned}$$

$$= 0.154141$$

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③

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$$\int_0^{1.8} \sqrt{1+x^2} \cdot dx = \int_0^{0.15} \sqrt{1+x^2} \cdot dx + \int_{0.15}^{0.30} \sqrt{1+x^2} \cdot dx + \dots + \int_{1.65}^{1.8} \sqrt{1+x^2} \cdot dx$$

$$\int_{0.30}^{0.45} \sqrt{1+x^2} \cdot dx \approx \frac{h}{2} \cdot [f(0.30) + f(0.45)]$$

$$= \frac{0.15}{2} \cdot [1.044031 + 1.096586]$$

$$= 0.160546$$

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$$\begin{aligned} \int_{0.45}^{0.60} \sqrt{1+x^2} \cdot dx &\approx \frac{h}{2} \cdot [f(0.45) + f(0.60)] \\ &= \frac{0.15}{2} \cdot [1.096586 + 1.166190] \end{aligned}$$

$$= 0.169708$$

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Order 1 $\rightarrow n = 1 \rightarrow$ The Trapezoidal Rule **12 times**

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$$\begin{aligned} \int_{0.60}^{0.75} \sqrt{1+x^2} \cdot dx &\approx \frac{h}{2} \cdot [f(0.60) + f(0.75)] \\ &= \frac{0.15}{2} \cdot [1.166190 + 1.250000] \end{aligned}$$

$$= 0.181214$$

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⑥

Order 1 $\rightarrow n = 1 \rightarrow$ The Trapezoidal Rule **12 times**

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$$\int_{0.75}^{0.90} \sqrt{1+x^2} \cdot dx \approx \frac{h}{2} \cdot [f(0.75) + f(0.90)]$$

$$= \frac{0.15}{2} \cdot [1.250000 + 1.345362]$$

$$= 0.194652$$

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Order 1 $\rightarrow n = 1 \rightarrow$ The Trapezoidal Rule **12 times**

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$$\begin{aligned} \int_{0.90}^{1.05} \sqrt{1+x^2} \cdot dx &\approx \frac{h}{2} \cdot [f(0.90) + f(1.05)] \\ &= \frac{0.15}{2} \cdot [1.345362 + 1.450000] \\ &= 0.209652 \end{aligned}$$

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⑧

Order 1 $\rightarrow n = 1 \rightarrow$ The Trapezoidal Rule 12 times

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$$\begin{aligned} \int_{1.05}^{1.20} \sqrt{1+x^2} \cdot dx &\approx \frac{h}{2} \cdot [f(1.05) + f(1.20)] \\ &= \frac{0.15}{2} \cdot [1.450000 + 1.562050] \\ &= 0.225904 \end{aligned}$$

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Order 1 $\rightarrow n = 1 \rightarrow$ The Trapezoidal Rule **12 times**

$$\int_0^{1.8} \sqrt{1+x^2} \cdot dx = \int_0^{0.15} \sqrt{1+x^2} \cdot dx + \int_{0.15}^{0.30} \sqrt{1+x^2} \cdot dx + \dots + \int_{1.65}^{1.8} \sqrt{1+x^2} \cdot dx$$

$$\begin{aligned} \int_{1.20}^{1.35} \sqrt{1+x^2} \cdot dx &\approx \frac{h}{2} \cdot [f(1.20) + f(1.35)] \\ &= \frac{0.15}{2} \cdot [1.562050 + 1.680030] \end{aligned}$$

$$= 0.243156$$

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Order 1 $\rightarrow n = 1 \rightarrow$ The Trapezoidal Rule 12 times

$$\int_0^{1.8} \sqrt{1+x^2} \cdot dx = \int_0^{0.15} \sqrt{1+x^2} \cdot dx + \int_{0.15}^{0.30} \sqrt{1+x^2} \cdot dx + \dots + \int_{1.65}^{1.8} \sqrt{1+x^2} \cdot dx$$

$$\int_{1.35}^{1.50} \sqrt{1+x^2} \cdot dx \approx \frac{h}{2} \cdot [f(1.35) + f(1.50)]$$
$$= \frac{0.15}{2} \cdot [1.680030 + 1.802776]$$

$$= 0.261210$$

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⑪

Order 1 $\rightarrow n = 1 \rightarrow$ The Trapezoidal Rule **12 times**

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$$\int_{1.50}^{1.65} \sqrt{1+x^2} \cdot dx \approx \frac{h}{2} \cdot [f(1.50) + f(1.65)]$$

$$= \frac{0.15}{2} \cdot [1.802776 + 1.929378]$$

$$= 0.279912$$

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Order 1 $\rightarrow n = 1 \rightarrow$ The Trapezoidal Rule **12 times**

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$$\int_{1.65}^{1.80} \sqrt{1+x^2} \cdot dx \approx \frac{h}{2} \cdot [f(1.65) + f(1.80)]$$

$$= \frac{0.15}{2} \cdot [1.929378 + 2.059126]$$

$$= 0.299138$$

⑫

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Order 1 → *n* = 1 → The Trapezoidal Rule 12 times

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$$\int_{0.00}^{1.80} \sqrt{1+x^2} \cdot dx \approx 0.150839 + 0.154141 + 0.160546 + 0.169708 + 0.181214 + 0.194652$$
$$+ 0.209652 + 0.225904 + 0.243156 + 0.261210 + 0.279912 + 0.299138$$
$$= 2.530073$$

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Order 2 $\rightarrow n = 2$

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12	1.80	2.059126

Order 2 $\rightarrow n = 2 \rightarrow$ Simpson's 1/3 Rule

Newton-Cotes Formulas of Closed Type

Example 06-01: Estimate the definite integral $\int_0^{1.8} \sqrt{1+x^2} \cdot dx$ using Newton-Cotes formulas of closed type of order 1,2,3 with 12 uniform subintervals.

i	x_i	$f(x_i)$
0	0.00	1.000000
1	0.15	1.011187
2	0.30	1.044031
3	0.45	1.096586
4	0.60	1.166190
5	0.75	1.250000
6	0.90	1.345362
7	1.05	1.450000
8	1.20	1.562050
9	1.35	1.680030
10	1.50	1.802776
11	1.65	1.929378
12	1.80	2.059126

Order 2 $\rightarrow n = 2 \rightarrow$ Simpson's 1/3 Rule 6 times

Newton-Cotes Formulas of Closed Type

Example 06-01: Estimate the definite integral $\int_0^{1.8} \sqrt{1+x^2} \cdot dx$ using Newton-Cotes formulas of closed type of order 1,2,3 with 12 uniform subintervals.

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4	0.60	1.166190
5	0.75	1.250000
6	0.90	1.345362
7	1.05	1.450000
8	1.20	1.562050
9	1.35	1.680030
10	1.50	1.802776
11	1.65	1.929378
12	1.80	2.059126

Order 2 $\rightarrow n = 2 \rightarrow$ Simpson's 1/3 Rule 6 times

$$\int_0^{1.8} \sqrt{1+x^2} \cdot dx = \int_0^{0.30} \sqrt{1+x^2} \cdot dx + \int_{0.30}^{0.60} \sqrt{1+x^2} \cdot dx + \dots + \int_{1.50}^{1.8} \sqrt{1+x^2} \cdot dx$$

Newton-Cotes Formulas of Closed Type

Example 06-01: Estimate the definite integral $\int_0^{1.8} \sqrt{1+x^2} \cdot dx$ using Newton-Cotes formulas of closed type of order 1,2,3 with 12 uniform subintervals.

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7	1.05	1.450000
8	1.20	1.562050
9	1.35	1.680030
10	1.50	1.802776
11	1.65	1.929378
12	1.80	2.059126

①

Order 2 $\rightarrow n = 2 \rightarrow$ Simpson's 1/3 Rule 6 times

$$\int_0^{1.8} \sqrt{1+x^2} \cdot dx = \int_0^{0.30} \sqrt{1+x^2} \cdot dx + \int_{0.30}^{0.60} \sqrt{1+x^2} \cdot dx + \dots + \int_{1.50}^{1.8} \sqrt{1+x^2} \cdot dx$$

$$\int_{0.00}^{0.30} \sqrt{1+x^2} \cdot dx \approx \frac{h}{3} \cdot [f(0.00) + 4 \cdot f(0.15) + f(0.30)]$$

$$= \frac{0.15}{3} \cdot [1.000000 + 4 \times 1.011187 + 1.044031]$$

$$= 0.304439$$

Newton-Cotes Formulas of Closed Type

Example 06-01: Estimate the definite integral $\int_0^{1.8} \sqrt{1+x^2} \cdot dx$ using Newton-Cotes formulas of closed type of order 1,2,3 with 12 uniform subintervals.

i	x_i	$f(x_i)$
0	0.00	1.000000
1	0.15	1.011187
2	0.30	1.044031
3	0.45	1.096586
4	0.60	1.166190
5	0.75	1.250000
6	0.90	1.345362
7	1.05	1.450000
8	1.20	1.562050
9	1.35	1.680030
10	1.50	1.802776
11	1.65	1.929378
12	1.80	2.059126

②

Order 2 $\rightarrow n = 2 \rightarrow$ Simpson's 1/3 Rule 6 times

$$\int_0^{1.8} \sqrt{1+x^2} \cdot dx = \int_0^{0.30} \sqrt{1+x^2} \cdot dx + \int_{0.30}^{0.60} \sqrt{1+x^2} \cdot dx + \dots + \int_{1.50}^{1.8} \sqrt{1+x^2} \cdot dx$$

$$\begin{aligned} \int_{0.30}^{0.60} \sqrt{1+x^2} \cdot dx &\approx \frac{h}{3} \cdot [f(0.30) + 4 \cdot f(0.45) + f(0.60)] \\ &= \frac{0.15}{3} \cdot [1.044031 + 4 \times 1.096586 + 1.166190] \end{aligned}$$

$$= 0.329828$$

Newton-Cotes Formulas of Closed Type

Example 06-01: Estimate the definite integral $\int_0^{1.8} \sqrt{1+x^2} \cdot dx$ using Newton-Cotes formulas of closed type of order 1,2,3 with 12 uniform subintervals.

i	x_i	$f(x_i)$
0	0.00	1.000000
1	0.15	1.011187
2	0.30	1.044031
3	0.45	1.096586
4	0.60	1.166190
5	0.75	1.250000
6	0.90	1.345362
7	1.05	1.450000
8	1.20	1.562050
9	1.35	1.680030
10	1.50	1.802776
11	1.65	1.929378
12	1.80	2.059126

③

Order 2 $\rightarrow n = 2 \rightarrow$ Simpson's 1/3 Rule 6 times

$$\int_0^{1.8} \sqrt{1+x^2} \cdot dx = \int_0^{0.30} \sqrt{1+x^2} \cdot dx + \int_{0.30}^{0.60} \sqrt{1+x^2} \cdot dx + \dots + \int_{1.50}^{1.8} \sqrt{1+x^2} \cdot dx$$

$$\int_{0.60}^{0.90} \sqrt{1+x^2} \cdot dx \approx \frac{h}{3} \cdot [f(0.60) + 4 \cdot f(0.75) + f(0.90)]$$

$$= \frac{0.15}{3} \cdot [1.166190 + 4 \times 1.250000 + 1.345362]$$

$$= 0.375578$$

Newton-Cotes Formulas of Closed Type

Example 06-01: Estimate the definite integral $\int_0^{1.8} \sqrt{1+x^2} \cdot dx$ using Newton-Cotes formulas of closed type of order 1,2,3 with 12 uniform subintervals.

i	x_i	$f(x_i)$
0	0.00	1.000000
1	0.15	1.011187
2	0.30	1.044031
3	0.45	1.096586
4	0.60	1.166190
5	0.75	1.250000
6	0.90	1.345362
7	1.05	1.450000
8	1.20	1.562050
9	1.35	1.680030
10	1.50	1.802776
11	1.65	1.929378
12	1.80	2.059126

④

Order 2 $\rightarrow n = 2 \rightarrow$ Simpson's 1/3 Rule 6 times

$$\int_0^{1.8} \sqrt{1+x^2} \cdot dx = \int_0^{0.30} \sqrt{1+x^2} \cdot dx + \int_{0.30}^{0.60} \sqrt{1+x^2} \cdot dx + \dots + \int_{1.50}^{1.8} \sqrt{1+x^2} \cdot dx$$

$$\begin{aligned} \int_{0.90}^{1.20} \sqrt{1+x^2} \cdot dx &\approx \frac{h}{3} \cdot [f(0.90) + 4 \cdot f(1.05) + f(1.20)] \\ &= \frac{0.15}{3} \cdot [1.345362 + 4 \times 1.450000 + 1.562050] \\ &= 0.435371 \end{aligned}$$

Newton-Cotes Formulas of Closed Type

Example 06-01: Estimate the definite integral $\int_0^{1.8} \sqrt{1+x^2} \cdot dx$ using Newton-Cotes formulas of closed type of order 1,2,3 with 12 uniform subintervals.

i	x_i	$f(x_i)$
0	0.00	1.000000
1	0.15	1.011187
2	0.30	1.044031
3	0.45	1.096586
4	0.60	1.166190
5	0.75	1.250000
6	0.90	1.345362
7	1.05	1.450000
8	1.20	1.562050
9	1.35	1.680030
10	1.50	1.802776
11	1.65	1.929378
12	1.80	2.059126

⑤

Order 2 $\rightarrow n = 2 \rightarrow$ Simpson's 1/3 Rule 6 times

$$\int_0^{1.8} \sqrt{1+x^2} \cdot dx = \int_0^{0.30} \sqrt{1+x^2} \cdot dx + \int_{0.30}^{0.60} \sqrt{1+x^2} \cdot dx + \dots + \int_{1.50}^{1.8} \sqrt{1+x^2} \cdot dx$$

$$\int_{1.20}^{1.50} \sqrt{1+x^2} \cdot dx \approx \frac{h}{3} \cdot [f(1.20) + 4 \cdot f(1.35) + f(1.50)]$$

$$= \frac{0.15}{3} \cdot [1.562050 + 4 \times 1.680030 + 1.802776]$$

$$= 0.504247$$

Newton-Cotes Formulas of Closed Type

Example 06-01: Estimate the definite integral $\int_0^{1.8} \sqrt{1+x^2} \cdot dx$ using Newton-Cotes formulas of closed type of order 1,2,3 with 12 uniform subintervals.

i	x_i	$f(x_i)$
0	0.00	1.000000
1	0.15	1.011187
2	0.30	1.044031
3	0.45	1.096586
4	0.60	1.166190
5	0.75	1.250000
6	0.90	1.345362
7	1.05	1.450000
8	1.20	1.562050
9	1.35	1.680030
10	1.50	1.802776
11	1.65	1.929378
12	1.80	2.059126

⑥

Order 2 $\rightarrow n = 2 \rightarrow$ Simpson's 1/3 Rule 6 times

$$\int_0^{1.8} \sqrt{1+x^2} \cdot dx = \int_0^{0.30} \sqrt{1+x^2} \cdot dx + \int_{0.30}^{0.60} \sqrt{1+x^2} \cdot dx + \dots + \int_{1.50}^{1.8} \sqrt{1+x^2} \cdot dx$$

$$\int_{1.50}^{1.80} \sqrt{1+x^2} \cdot dx \approx \frac{h}{3} \cdot [f(1.50) + 4 \cdot f(1.65) + f(1.80)]$$

$$= \frac{0.15}{3} \cdot [1.802776 + 4 \times 1.929378 + 2.059126]$$

$$= 0.578971$$

Newton-Cotes Formulas of Closed Type

Example 06-01: Estimate the definite integral $\int_0^{1.8} \sqrt{1+x^2} \cdot dx$ using Newton-Cotes formulas of closed type of order 1,2,3 with 12 uniform subintervals.

i	x_i	$f(x_i)$
0	0.00	1.000000
1	0.15	1.011187
2	0.30	1.044031
3	0.45	1.096586
4	0.60	1.166190
5	0.75	1.250000
6	0.90	1.345362
7	1.05	1.450000
8	1.20	1.562050
9	1.35	1.680030
10	1.50	1.802776
11	1.65	1.929378
12	1.80	2.059126

Order 2 $\rightarrow n = 2 \rightarrow$ Simpson's 1/3 Rule 6 times

$$\int_0^{1.8} \sqrt{1+x^2} \cdot dx = \int_0^{0.30} \sqrt{1+x^2} \cdot dx + \int_{0.30}^{0.60} \sqrt{1+x^2} \cdot dx + \dots + \int_{1.50}^{1.8} \sqrt{1+x^2} \cdot dx$$

$$\int_{0.00}^{1.80} \sqrt{1+x^2} \cdot dx \approx 0.304439 + 0.329828 + 0.375578 + 0.435371 + 0.504247 + 0.578971$$

$= 2.528433$

Newton-Cotes Formulas of Closed Type

Example 06-01: Estimate the definite integral $\int_0^{1.8} \sqrt{1+x^2} \cdot dx$ using Newton-Cotes formulas of closed type of order 1,2,3 with 12 uniform subintervals.

i	x_i	$f(x_i)$
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4	0.60	1.166190
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7	1.05	1.450000
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Newton-Cotes Formulas of Closed Type

Example 06-01: Estimate the definite integral $\int_0^{1.8} \sqrt{1+x^2} \cdot dx$ using Newton-Cotes formulas of closed type of order 1,2,3 with 12 uniform subintervals.

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10	1.50	1.802776
11	1.65	1.929378
12	1.80	2.059126

Order 3 $\rightarrow n = 3$

Newton-Cotes Formulas of Closed Type

Example 06-01: Estimate the definite integral $\int_0^{1.8} \sqrt{1+x^2} \cdot dx$ using Newton-Cotes formulas of closed type of order 1,2,3 with 12 uniform subintervals.

i	x_i	$f(x_i)$
0	0.00	1.000000
1	0.15	1.011187
2	0.30	1.044031
3	0.45	1.096586
4	0.60	1.166190
5	0.75	1.250000
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7	1.05	1.450000
8	1.20	1.562050
9	1.35	1.680030
10	1.50	1.802776
11	1.65	1.929378
12	1.80	2.059126

Order 3 $\rightarrow n = 3 \rightarrow$ Simpson's 3/8 Rule

Newton-Cotes Formulas of Closed Type

Example 06-01: Estimate the definite integral $\int_0^{1.8} \sqrt{1+x^2} \cdot dx$ using Newton-Cotes formulas of closed type of order 1,2,3 with 12 uniform subintervals.

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2	0.30	1.044031
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4	0.60	1.166190
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11	1.65	1.929378
12	1.80	2.059126

Order 3 → $n = 3$ → Simpson's 3/8 Rule 4 times

Newton-Cotes Formulas of Closed Type

Example 06-01: Estimate the definite integral $\int_0^{1.8} \sqrt{1+x^2} \cdot dx$ using Newton-Cotes formulas of closed type of order 1,2,3 with 12 uniform subintervals.

i	x_i	$f(x_i)$
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7	1.05	1.450000
8	1.20	1.562050
9	1.35	1.680030
10	1.50	1.802776
11	1.65	1.929378
12	1.80	2.059126

Order 3 $\rightarrow n = 3 \rightarrow$ Simpson's 3/8 Rule 4 times

$$\int_0^{1.8} \sqrt{1+x^2} \cdot dx = \int_0^{0.45} \sqrt{1+x^2} \cdot dx + \int_{0.45}^{0.90} \sqrt{1+x^2} \cdot dx + \dots + \int_{1.35}^{1.8} \sqrt{1+x^2} \cdot dx$$

Newton-Cotes Formulas of Closed Type

Example 06-01: Estimate the definite integral $\int_0^{1.8} \sqrt{1+x^2} \cdot dx$ using Newton-Cotes formulas of closed type of order 1,2,3 with 12 uniform subintervals.

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5	0.75	1.250000
6	0.90	1.345362
7	1.05	1.450000
8	1.20	1.562050
9	1.35	1.680030
10	1.50	1.802776
11	1.65	1.929378
12	1.80	2.059126

①

Order 3 $\rightarrow n = 3 \rightarrow$ Simpson's 3/8 Rule 4 times

$$\int_0^{1.8} \sqrt{1+x^2} \cdot dx = \int_0^{0.45} \sqrt{1+x^2} \cdot dx + \int_{0.45}^{0.90} \sqrt{1+x^2} \cdot dx + \dots + \int_{1.35}^{1.8} \sqrt{1+x^2} \cdot dx$$

$$\begin{aligned} \int_{0.00}^{0.45} \sqrt{1+x^2} \cdot dx &\approx \frac{3h}{8} \cdot [f(0.00) + 3 \cdot f(0.15) + 3 \cdot f(0.30) + f(0.45)] \\ &= \frac{3 \times 0.15}{8} \cdot [1.000000 + 3 \times 1.011187 + 3 \times 1.044031 + 1.096586] \end{aligned}$$

$$= 0.464751$$

Newton-Cotes Formulas of Closed Type

Example 06-01: Estimate the definite integral $\int_0^{1.8} \sqrt{1+x^2} \cdot dx$ using Newton-Cotes formulas of closed type of order 1,2,3 with 12 uniform subintervals.

i	x_i	$f(x_i)$
0	0.00	1.000000
1	0.15	1.011187
2	0.30	1.044031
3	0.45	1.096586
4	0.60	1.166190
5	0.75	1.250000
6	0.90	1.345362
7	1.05	1.450000
8	1.20	1.562050
9	1.35	1.680030
10	1.50	1.802776
11	1.65	1.929378
12	1.80	2.059126

②

Order 3 $\rightarrow n = 3 \rightarrow$ Simpson's 3/8 Rule 4 times

$$\int_0^{1.8} \sqrt{1+x^2} \cdot dx = \int_0^{0.45} \sqrt{1+x^2} \cdot dx + \int_{0.45}^{0.90} \sqrt{1+x^2} \cdot dx + \dots + \int_{1.35}^{1.8} \sqrt{1+x^2} \cdot dx$$

$$\begin{aligned} \int_{0.45}^{0.90} \sqrt{1+x^2} \cdot dx &\approx \frac{3h}{8} \cdot [f(0.45) + 3 \cdot f(0.60) + 3 \cdot f(0.75) + f(0.90)] \\ &= \frac{3 \times 0.15}{8} \cdot [1.096586 + 3 \times 1.166190 + 3 \times 1.250000 + 1.345362] \end{aligned}$$

$$= 0.545092$$

Newton-Cotes Formulas of Closed Type

Example 06-01: Estimate the definite integral $\int_0^{1.8} \sqrt{1+x^2} \cdot dx$ using Newton-Cotes formulas of closed type of order 1,2,3 with 12 uniform subintervals.

i	x_i	$f(x_i)$
0	0.00	1.000000
1	0.15	1.011187
2	0.30	1.044031
3	0.45	1.096586
4	0.60	1.166190
5	0.75	1.250000
6	0.90	1.345362
7	1.05	1.450000
8	1.20	1.562050
9	1.35	1.680030
10	1.50	1.802776
11	1.65	1.929378
12	1.80	2.059126

③

Order 3 $\rightarrow n = 3 \rightarrow$ Simpson's 3/8 Rule 4 times

$$\int_0^{1.8} \sqrt{1+x^2} \cdot dx = \int_0^{0.45} \sqrt{1+x^2} \cdot dx + \int_{0.45}^{0.90} \sqrt{1+x^2} \cdot dx + \dots + \int_{1.35}^{1.8} \sqrt{1+x^2} \cdot dx$$

$$\begin{aligned} \int_{0.90}^{1.35} \sqrt{1+x^2} \cdot dx &\approx \frac{3h}{8} \cdot [f(0.90) + 3 \cdot f(1.05) + 3 \cdot f(1.20) + f(1.35)] \\ &= \frac{3 \times 0.15}{8} \cdot [1.345362 + 3 \times 1.450000 + 3 \times 1.562050 + 1.680030] \end{aligned}$$

$$= 0.678462$$

Newton-Cotes Formulas of Closed Type

Example 06-01: Estimate the definite integral $\int_0^{1.8} \sqrt{1+x^2} \cdot dx$ using Newton-Cotes formulas of closed type of order 1,2,3 with 12 uniform subintervals.

i	x_i	$f(x_i)$
0	0.00	1.000000
1	0.15	1.011187
2	0.30	1.044031
3	0.45	1.096586
4	0.60	1.166190
5	0.75	1.250000
6	0.90	1.345362
7	1.05	1.450000
8	1.20	1.562050
9	1.35	1.680030
10	1.50	1.802776
11	1.65	1.929378
12	1.80	2.059126

4

Order 3 $\rightarrow n = 3 \rightarrow$ Simpson's 3/8 Rule 4 times

$$\int_0^{1.8} \sqrt{1+x^2} \cdot dx = \int_0^{0.45} \sqrt{1+x^2} \cdot dx + \int_{0.45}^{0.90} \sqrt{1+x^2} \cdot dx + \dots + \int_{1.35}^{1.8} \sqrt{1+x^2} \cdot dx$$

$$\begin{aligned} \int_{1.35}^{1.80} \sqrt{1+x^2} \cdot dx &\approx \frac{3h}{8} \cdot [f(1.35) + 3 \cdot f(1.50) + 3 \cdot f(1.65) + f(1.80)] \\ &= \frac{3 \times 0.15}{8} \cdot [1.680030 + 3 \times 1.802776 + 3 \times 1.929378 + 2.059126] \end{aligned}$$

$$= 0.840128$$

Newton-Cotes Formulas of Closed Type

Example 06-01: Estimate the definite integral $\int_0^{1.8} \sqrt{1+x^2} \cdot dx$ using Newton-Cotes formulas of closed type of order 1,2,3 with 12 uniform subintervals.

i	x_i	$f(x_i)$
0	0.00	1.000000
1	0.15	1.011187
2	0.30	1.044031
3	0.45	1.096586
4	0.60	1.166190
5	0.75	1.250000
6	0.90	1.345362
7	1.05	1.450000
8	1.20	1.562050
9	1.35	1.680030
10	1.50	1.802776
11	1.65	1.929378
12	1.80	2.059126

Order 3 $\rightarrow n = 3 \rightarrow$ Simpson's 3/8 Rule 4 times

$$\int_0^{1.8} \sqrt{1+x^2} \cdot dx = \int_0^{0.45} \sqrt{1+x^2} \cdot dx + \int_{0.45}^{0.90} \sqrt{1+x^2} \cdot dx + \dots + \int_{1.35}^{1.8} \sqrt{1+x^2} \cdot dx$$

$$\int_{0.00}^{1.80} \sqrt{1+x^2} \cdot dx \approx 0.464751 + 0.545092 + 0.678462 + 0.840128$$

$$= 2.528433$$

Newton-Cotes Formulas of Closed Type

Newton-Cotes Formulas of Closed Type

Example 06-02: Estimate the definite integral from 0.30 to 0.95 of the following data using Newton-Cotes formulas of closed type of order 1 and 2.

x_i	$f(x_i)$
0.30	1.04403
0.40	1.07703
0.65	1.19269
0.85	1.31244
0.95	1.37931

Assignment 4

1. Approximate $f'(0.65)$, $f''(0.65)$, $f^{(3)}(0.65)$ using the data from 0.20 to 0.80.
2. Estimate definite integral of $f(x)$ using Newton-Cotes formulas of closed type of order 1,2,3 from the provided data set.

i	x_i	$f(x_i)$
0	0.20	1.266491
1	0.35	1.439249
2	0.50	1.600781
3	0.65	1.757075
4	0.80	1.912067
5	0.95	2.068499
6	1.10	2.228340
7	1.25	2.393024
8	1.40	2.563591
9	1.55	2.740791
10	1.70	2.925150
11	1.85	3.117020
12	2.00	3.316625

